

A Narrator Pipeline driven by DMLG

Structural Hybrid Intelligence through an IEGR-Complete Generative System

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Abstract

This paper presents a structural perspective on how advanced artificial systems may organize and behave when approaching general-purpose intelligence. Rather than proposing a concrete realization of AGI, we introduce the IEGR framework—energetic efficiency (E), informational richness (I), geometric structure (G), and recursive depth (R)—as a conceptual lens for understanding how complementary architectures can be composed into coherent, recursively stable generative systems.

Within this framework, we describe DMLG-v3, a deterministic geometric-recursive substrate designed for world absorption and long-horizon narrative coherence. DMLG-v3 provides canonical grammar, modular page structure, stable attractors, and recursive memory, enabling it to internalize and reproduce entire fictional worlds. Coupled with a transformer-based LLM supplying energetic and informational roles, and integrated through a bi-directional narrator pipeline, the resulting hybrid system functions as an example of how IEGR-complete behavior may emerge through structural composition rather than scale alone.

Empirically, the narrator pipeline autonomously generates multi-chapter works—including *Tales of the Dark Forest* and *Tales of the Shrouded City*—demonstrating stable attractors, strict canonical constraints, and coherent long-form structure across dozens of chapters. These results illustrate how hybrid recursion between complementary model types can yield systems capable of absorbing domains, maintaining internal canon, and producing multi-generational derivative narratives.

The key insight of this work is that IEGR-completeness provides a structural template for understanding potential AGI-like behavior without asserting its realization. Hybrid architectures, when equipped with scalable energetic capacity, may serve as practical examples of how general-purpose reasoning, world absorption, and recursive narrative evolution could emerge in future systems. The paper concludes with structural predictions and open questions concerning energetic scaling laws, domain generalization, and the quantitative geometry of IEGR-scale.

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1 Introduction

Artificial intelligence has advanced rapidly over the past decade, yet the field still lacks a structural account of how general-purpose intelligent behavior may emerge. Existing discussions of artificial general intelligence (AGI) tend to be anthropocentric or task-based, focusing on what AGI should *do* rather than on the architectural conditions under which broad, coherent behavior could arise. As a result, debates about AGI remain fragmented, oscillating between scale-driven optimism, modular skepticism, and safety concerns that lack a unifying structural framework.

This paper approaches the question from a different angle. Rather than proposing a concrete realization of AGI, we introduce the IEGR framework—energetic efficiency (E), informational richness (I), geometric structure (G), and recursive depth (R)—as a conceptual lens for understanding how complementary systems may be combined to produce stable, long-horizon behavior. IEGR is substrate-agnostic: it applies equally to biological cognition, artificial architectures, and hybrid systems. The framework does not define AGI; instead, it identifies the structural roles that must be present for general-purpose reasoning to be possible.

Modern AI systems instantiate these roles asymmetrically. Transformer-based large language models (LLMs) scale E and I aggressively but lack geometric modularity and recursive depth, placing them in the *statistical parrot* regime. Conversely, structured modular systems such as the Dynamic Modular Language Graph (DMLG) instantiate G and R strongly but lack large-scale E and I, placing them in the *canonical parrot* regime. Neither architecture alone provides the full IEGR spectrum, but together they form a complementary hybrid capable of coherent long-horizon behavior.

To explore this structural complementarity, we introduce DMLG-v3, a deterministic geometric-recursive substrate designed for world absorption, canonical grammar, stable attractors, and recursive memory. When coupled with a transformer-based LLM and integrated through a bi-directional narrator pipeline, the resulting system serves as an example of how IEGR-complete behavior can emerge through hybrid recursion rather than scale alone. The narrator pipeline is not presented as AGI; instead, it functions as a concrete demonstration of how AGI-like structural properties—world internalization, multi-generational narrative evolution, and domain-stable behavior—may arise in practice.

Empirically, the pipeline autonomously generates long-form narratives that maintain internal canon across dozens of chapters. Two complete works—*Tales of the Dark Forest* and *Tales of the Shrouded City*—illustrate how hybrid recursion between complementary model types can yield systems capable of absorbing domains, maintaining stylistic and structural coherence, and producing derivative narratives across multiple editions.

The contributions of this paper are fourfold:

1. We formalize the IEGR framework as a structural model for analyzing complementary roles in intelligent systems.
2. We introduce DMLG-v3 as a deterministic geometric-recursive substrate suitable for hybrid generative architectures.
3. We present the sequential narrator pipeline as a concrete example of IEGR-complete hybrid behavior.
4. We provide empirical evidence through two autonomous narrative universes and outline structural predictions for future hybrid systems.

Together, these results demonstrate how complementary parrot regimes can be composed into structurally coherent systems, and how hybrid architectures may serve as practical examples of AGI-like behavior without asserting AGI as a realized capability.

2 Motivation

The motivation for this work is both conceptual and empirical. Conceptually, the goal is to explore how complementary artificial systems can be composed into structures that exhibit coherent long-horizon behavior. Rather than proposing a concrete realization of AGI, this work investigates how the IEGR framework—energetic efficiency (E), informational richness (I), geometric structure (G), and recursive depth (R)—can serve as a structural lens for understanding hybrid architectures capable of world absorption, canonical constraint enforcement, and recursive narrative evolution.

Empirically, the goal is to demonstrate that a hybrid system composed of complementary parrot regimes—DMLG-v3 and a transformer-based LLM—can exceed the long-horizon narrative capabilities of prompt-engineered LLMs and produce autonomous, canon-consistent fictional worlds. The sequential narrator pipeline serves as a concrete example of how structured recursion between heterogeneous models can yield stable, domain-bound intelligent behavior.

2.1 DMLG as a Geometric–Recursive Substrate

Within the IEGR framework, modern LLMs instantiate energetic and informational roles strongly but lack geometric modularity and recursive depth. Conversely, DMLG-v3 instantiates G and R through canonical grammar, modular page structure, stable attractors, and long-horizon narrative memory. This complementarity motivates the use of DMLG-v3 as a *generic geometric-recursive substrate* within hybrid systems.

DMLG-v3 provides:

- canonical geometry through prefix grammar and modular pages,
- recursive depth through narrative memory and context accumulation,
- deterministic structure suitable for inspection and governance,
- plasticity through curriculum learning and world absorption.

These properties make DMLG-v3 suitable for hybrid architectures that require structured recursion, stable world models, and long-horizon coherence.

2.2 Domain-Bound Intelligent Behavior

A second motivation is to illustrate how hybrid systems can exhibit *domain-bound intelligent behavior*: coherent reasoning, recursive planning, and multi-generational world evolution within a constrained fictional universe. The narrator pipeline demonstrates that hybrid recursion can achieve:

- autonomous world absorption,
- multi-generational narrative evolution,
- strict canonical constraint enforcement,
- emergent behavioral attractors,
- long-term memory and recursive state tracking.

These capabilities exceed what current LLMs can achieve through prompt engineering alone and highlight the importance of structured GR substrates.

2.3 Beyond Prompt Engineering: Structured Long-Horizon Narration

Prompt-engineered LLMs struggle with canonical consistency, world memory, and long-horizon narrative coherence. They exhibit structural instability due to missing G and R roles. The narrator pipeline demonstrates that:

- DMLG-v3 stabilizes narrative geometry,
- the pipeline provides recursive moderation and planning,
- the LLM supplies energetic and informational amplification,
- the hybrid system maintains coherence across entire books.

This shows that hybrid architectures can surpass the narrative capabilities of pure transformer systems and provide a path toward structured long-form generation.

2.4 End-to-End Autonomous AI Generation

A core motivation is to demonstrate *end-to-end autonomous AI generation*: a system that produces entire books without human intervention in content. The sequential narrator pipeline achieves this through:

- DMLG seed generation,
- LLM moderation and scoring,
- recursive keyword evolution,
- context accumulation and narrative memory,
- automatic transitions and ending validation.

The resulting works—*Tales of the Dark Forest* and *Tales of the Shrouded City*—illustrate how hybrid systems can maintain internal canon and produce coherent long-form narratives autonomously.

2.5 Canon-Consistent and Domain-Bound Generation

Another motivation is to demonstrate that hybrid systems can maintain strict canonical constraints within a fictional universe. DMLG-v3’s geometric structure and recursive memory enable:

- consistent character behavior,
- stable thematic evolution,
- domain-specific physics and stylistic rules,
- prohibition of out-of-domain motifs,
- multi-generational world coherence.

This positions DMLG-v3 as a substrate for *canon-bound artificial cognition* within constrained narrative domains.

2.6 IEGR-Complete Hybrid Systems

Finally, the motivation is to explore how hybrid systems can instantiate the structural profile of IEGR-complete behavior: systems in which all four roles—E, I, G, and R—are present and functionally integrated. Such systems do not constitute AGI, but they provide a structural hint of how AGI-like behavior may emerge through architectural complementarity rather than scale alone.

2.7 Summary

The motivation for this work is to show that:

- DMLG-v3 can serve as a geometric-recursive substrate,
- hybrid systems can exhibit domain-bound intelligent behavior,
- structured pipelines can exceed LLM prompt engineering,
- end-to-end autonomous AI generation is feasible,
- canon-consistent world evolution is achievable,
- IEGR-complete behavior can emerge through hybrid recursion.

The sequential narrator pipeline provides a fully documented example of such a system operating at narrative scale.

3 Related Work

Research on autonomous narration, hybrid modular architectures, and dynamically evolving generative systems spans several partially connected domains. The present work builds on three strands of prior research: (1) the structural foundations of the *Triadic Cosmos* ecosystem, (2) iterative and diffusion-style generative methodologies, and (3) emerging work on AI life, self-recursive learning loops, and dynamically evolving cognitive systems.

3.1 Triadic Cosmos Foundations

The Dynamic Modular Language Graph (DMLG) Witte 2026c introduced a deterministic, graph-based governance architecture in which language generation is modeled as traversal through a canonical token-transition graph. DMLG provides the geometric (G) and recursive (R) substrate for the narrator pipeline: explicit structure, modular paging, deterministic transitions, and PRE-based governance. Its emphasis on structure over scale directly motivates the canonical-parrot regime in the IEGR taxonomy.

Dynamic Modular Language Models (DMLMs) Witte 2026d form the predecessor architecture to DMLG: a multi-agent, evaluator-governed system in which micro-models, paging structures, and a PRE loop produce coherent long-form narrative structure without transformers or large-scale training. Although DMLMs focus on generation rather than governance, they establish the architectural principles—modularity, explicit interfaces, evaluator-driven recursion—that underpin the narrator pipeline’s raw generative substrate.

The structural taxonomy of intelligence (IEGR) Witte 2026a provides the conceptual framework for understanding hybrid systems as *sophisticated parrots*: systems that are IEGR-complete but not yet IEGR-scaled. In this taxonomy, LLMs supply energetic and informational roles (E, I), while DMLG supplies geometric and recursive roles (G, R). The narrator pipeline is therefore positioned as a concrete instantiation of IEGR complementarity: a hybrid system whose long-form generative capability arises from the integration of complementary structural roles.

The diffusion-driven narration framework Witte 2026b introduced the idea that long-form narrative structure can emerge through staged denoising across heterogeneous cognitive modules. In this methodology, DMLG provides canonical geometry and recursive scaffolding, while LLMs supply semantic refinement, evaluative judgment, and informational expansion. The resulting system behaves as a distributed cognitive architecture rather than a single generative model.

The present work does not replace or extend the full diffusion-driven narration stack. Instead, it introduces DMLG-v3 as a new geometric-recursive pillar for the sequential narrator pipeline. In this pipeline, only the first diffusion stage is performed: a structured DMLG-v3 generation followed by refinement through a mid-scale LLM (e.g. Mistral). Later diffusion stages—large-LLM remastering and the Director’s Cut—remain part of the broader diffusion-driven narration paradigm and were applied in the empirical manuscripts, but they are not part of the pipeline analyzed in this paper.

This positions the present work as a detailed examination of the DMLG-v3 substrate and the sequential narrator pipeline within the larger framework of diffusion-driven narration. The paper focuses on how DMLG-v3 enables stable world absorption, canonical grammar, and long-horizon recursion, and how these properties interact with a lightweight LLM layer to produce autonomous, canon-consistent narrative generations.

3.2 Narrative Pipelines and Autonomous Story Generation

Narrative generation has been explored in several domains, though most systems lack the structural completeness required for long-horizon coherence. Early work on story grammars and plot machines (e.g., Proppian morphology, M. O. Riedl and Young 2010) introduced symbolic pipelines but lacked semantic richness (I) and energetic capacity (E). Neural story generators such as `PlotMachines` Yao et al. 2019 and hierarchical planners Ammanabrolu and M. Riedl 2020 improved semantic fluency but remained structurally shallow, relying on transformer-based architectures without geometric modularity (G) or recursive depth (R).

Interactive fiction agents (e.g., `Jericho` Hausknecht et al. 2020) demonstrated multi-step reasoning in constrained environments but lacked long-term narrative memory and world absorption. Recent work on LLM-driven story generation (e.g., `SIL0` Chang et al. 2023, `Re3` Shinn, Cassano, et al. 2023) introduced reflective loops and self-correction, but these systems remain statistical parrots: high E and I, deficient G and R, and structurally unstable over long horizons.

The sequential narrator pipeline differs fundamentally from these approaches. Rather than relying on transformer recursion, it couples a geometric-recursive substrate (DMLG-v3) with an energetic-informational substrate (LLM) through a bi-directional IEGR loop. This enables:

- autonomous world absorption,
- multi-generational narrative evolution,
- strict canonical constraint enforcement,
- emergent behavioral attractors,
- long-horizon coherence across dozens of chapters.

To our knowledge, no prior narrative system demonstrates these properties.

3.3 Hybrid Cognitive Architectures and Self-Recursive Systems

Hybrid cognitive systems have been explored in several contexts. Program-induction architectures such as `DreamCoder` Ellis et al. 2021 demonstrate recursive self-improvement but lack

semantic breadth (I). Neural module networks Andreas et al. 2016 provide geometric structure (G) but lack recursive depth (R) and energetic capacity (E). Reflective LLM systems such as **Reflexion** Shinn, Cassano, et al. 2023 and **Self-Refine** Madaan et al. 2023 introduce shallow recursion but remain structurally unstable due to missing geometric constraints.

Work on “AI life” and self-evolving agents (e.g., Stanley, Lehman, and Soros 2017, Wang et al. 2024) explores open-endedness but lacks canonical governance and long-horizon narrative structure. The narrator pipeline provides a different perspective: self-recursion is not emergent chaos but a governed IEGR-complete loop in which each structural role is explicitly instantiated.

3.4 Positioning of the Present Work

The present work differs from prior literature in three ways:

1. It provides a **structurally complete hybrid architecture** in which each IEGR role is explicitly instantiated and integrated.
2. It demonstrates **autonomous world absorption and multi-generational narrative evolution**, including two full books— *Tales of the Dark Forest* and *Tales of the Shrouded City*—as empirical evidence.
3. It positions narrative generation not as a stylistic task but as a **testbed for AGI**: a domain requiring long-horizon coherence, recursive planning, canonical consistency, and dynamic world evolution.

In this context, the sequential narrator pipeline serves as the first fully documented example of an IEGR-complete hybrid system operating at narrative scale. It demonstrates that complementary parrot regimes can be composed into structurally complete intelligent systems, and that energetic scaling provides a clear path from sophisticated parrots to human-like AGI.

4 Fixed Output Size and Codebook-Based Deterministic Decoding

A central innovation of DMLG-v3 is the introduction of a *fixed-dimensional* output space combined with a hybrid one-hot and triple-hot codebook. Unlike DMLG-v2, where output dimensionality and token embeddings drifted during training, DMLG-v3 enforces a compile-time constant output dimension D . Each token is mapped to a codebook vector inside the immutable output space.

The codebook partitions the output dimension into:

$$D_{\text{one-hot}} = D/4, \quad D_{\text{triple-hot}} = 3D/4,$$

where the first $D/4$ positions correspond to one-hot tokens and the remaining $3D/4$ positions form the triple-hot combinatorial space. The assignment is *page-local*: each page determines which tokens occupy the one-hot region and which occupy the triple-hot region, based on the page’s internal vocabulary ordering. Tokens with low page-local indices receive one-hot codes, while higher indices receive triple-hot codes.

Triple-hot tokens are encoded by selecting three distinct positions from the $D_{\text{triple-hot}}$ region. The number of available triple-hot codes is:

$$N_{\text{triple-hot}} = \binom{D_{\text{triple-hot}}}{3}.$$

Two concrete examples illustrate the scalability of this scheme:

Example 1: $D = 32$

$$D_{\text{one-hot}} = 8, \quad D_{\text{triple-hot}} = 24.$$

The triple-hot capacity is:

$$\binom{24}{3} = \frac{24 \cdot 23 \cdot 22}{6} = 2024.$$

Thus a 32-dimensional output space supports:

- 8 one-hot tokens,
- 2024 triple-hot tokens,
- for a total of 2032 page-local tokens.

For typical single-book curricula, this is more than sufficient: most pages require fewer than 64 tokens, and many pages fit entirely within the one-hot region.

Example 2: $D = 256$

$$D_{\text{one-hot}} = 64, \quad D_{\text{triple-hot}} = 192.$$

The triple-hot capacity is:

$$\binom{192}{3} = \frac{192 \cdot 191 \cdot 190}{6} = 1,161,280.$$

This supports:

- 64 one-hot tokens,
- over 1.16 million triple-hot tokens,
- enabling extremely large page-local vocabularies.

This scale is suitable for multi-book curricula, multi-world absorption, or high-entropy generative regimes.

This hybrid codebook constitutes a structural breakthrough: informational richness (I) scales independently of energetic cost (E) and geometric structure (G). The model becomes a deterministic substrate capable of absorbing entire fictional universes while maintaining canonical decoding. In the IEGR taxonomy, the fixed output space supplies the geometric backbone required for stable recursive behavior and long-horizon narrative coherence.

5 Prefix Grammar and Deterministic Grammar–Lemma Alternation

DMLG-v3 replaces the unstable postfix grammar of V2 with a fully prefix-based canonical grammar. Every non-terminal token is represented as:

<TAG> lemma,

while terminal tokens such as <PERIOD> and <EOL> carry no lemma. This enforces a strict alternation between grammar tokens (structure) and lemma tokens (content), encoded explicitly via a grammar flag in the input stream.

The resulting update loop follows the triadic IEGR cycle:

$$E \rightarrow G \rightarrow I,$$

where energy selects the next token, geometry constrains its canonical form, and information is supplied by the lemma. Recursion (R) emerges from repeated applications of this loop across the context window and narrative memory.

Prefix grammar stabilizes category detection, eliminates syntactic drift, and provides the geometric structure (G) that transformer-based systems lack. Lemma alternation becomes deterministic rather than emergent, ensuring that narrative sequences remain canonical across long horizons. This redesign is a structural upgrade rather than a cosmetic change: it enables DMLG-v3 to function as a genuine G–R engine within IEGR-complete hybrid systems.

6 Additional DMLG Architectural Improvements

Beyond the two core innovations above, DMLG-v3 introduces a series of structural refinements that collectively eliminate V2’s instability and transform the model into a deterministic canonical substrate.

Fixed Neural Architecture

All page-level MLPs in DMLG-v3 share a single, fixed-size architecture defined at compile time. Hidden sizes, activation sizes, and output dimensions are immutable throughout training. This eliminates the dynamic resizing behaviour present in DMLG-v2, preventing structural drift and ensuring full reproducibility across epochs, curricula, and narrative domains.

Each page MLP is therefore an instance of the same canonical update operator:

$$x \rightarrow h_1 \rightarrow h_2 \rightarrow y,$$

with identical layer widths and identical parameterization. Only the page-local weights differ, not the architecture itself. This fixed structure provides a stable geometric backbone for recursive updates across pages, context windows, and narrative memory.

Hybrid SiLU→GELU Activation Stack

All page MLPs in DMLG-v3 share the same fixed-size activation MLP, identical in width and structure to the static design used in DMLG-v2. What changes in v3 is the internal nonlinearity: the two hidden layers use different activation functions, forming a lightweight hybrid stack.

$$h_1 = \text{SiLU}(W_1x + b_1), \quad h_2 = \text{GELU}(W_2h_1 + b_2).$$

The first hidden layer applies **SiLU**, providing smooth curvature and stable early-stage structural shaping. The second hidden layer applies **GELU**, adding transformer-grade expressivity and finer semantic resolution. This hybrid design replaces the earlier SiLU+GELU mixed activation used in v2 and yields a clearer separation between structural stability (SiLU) and expressive refinement (GELU).

Because the activation MLP is shared across all pages and fixed at compile time, every page MLP benefits from the same stable geometry and expressive refinement, without architectural drift or parameter growth.

Warmup-First Training Regime

DMLG-v3 removes the exploration phase of V2 and replaces it with a strictly ordered *warmup-first* training regime. Warmup epochs present the curriculum in its canonical sequence order: every page is trained in the exact order in which it appears in the book. This stabilizes grammar, page geometry, token transitions, and the attractor structure before any stochastic sampling occurs.

After warmup, the model switches to *random epochs*, where curriculum sequences are sampled in random order. These epochs refine semantic clusters and local stylistic variation without altering the underlying structural geometry established during warmup.

1. **Warmup epochs (sequential curriculum).** Sequences are shown in deterministic book order, allowing the model to learn stable grammar, canonical page structure, and attractor geometry.
2. **Random epochs (stochastic curriculum).** Sequences are sampled uniformly at random, refining semantics and local variation while preserving the fixed structural backbone learned during warmup.

This warmup-first regime yields early convergence to stable attractors and prevents structural drift, enabling DMLG-v3 to absorb entire narrative worlds with consistent geometry and long-horizon coherence.

Lemma-Keyed Pages and Transition Map Removal

Pages remain lemma-keyed as in V2, but V3 introduces fixed page counts, fixed page MLP sizes, stable embeddings, and removes the transition map entirely. Transitions are now learned implicitly through recursive page updates, improving geometric modularity and scalability.

Stabilized Context Window and Narrative Memory

Deterministic grammar and fixed embeddings make narrative memory fully functional. Actors, actions, and atmosphere are tracked reliably across long sequences, supplying recursive depth (R).

Modern Decoding Stack

DMLG-v3 uses a modern decoding stack combining top- k sampling, temperature scaling, grammar-constrained token selection, and beam search with jitter and damping. Beam search operates as the primary decoding method but can fall back to stochastic sampling when no valid continuation exists, preventing dead ends and maintaining generative flow.

Semantic validation filters structurally inconsistent or drift-prone candidates, ensuring that only grammar-respecting and canon-consistent continuations enter the recursive loop. Together, these mechanisms provide stable long-horizon recursion and prevent attractor drift during sequential narration.

7 The Sequential Narrator Pipeline

The sequential narrator pipeline is a bi-directional hybrid intelligence system that integrates DMLG-v3 as a geometric-recursive substrate, a transformer-based LLM as an energetic-informational substrate, and a recursive coupling layer that binds both components into an IEGR-complete architecture. Unlike earlier non-sequential pipelines, the sequential variant maintains a persistent narrative context, evolves keyword sets dynamically, and constructs entire books chapter by chapter through long-horizon recursive updates.

This section formalizes the architecture and operational flow of the sequential pipeline within the broader diffusion-driven narration paradigm.

7.1 Pipeline Overview

The pipeline consists of three tightly coupled functional components:

1. **Structural Generator (DMLG-v3)** Produces canonical seed stories using prefix grammar, lemma-keyed pages, narrative memory, and modern decoding strategies.
2. **LLM Evaluator Layer** Moderates, validates, scores, titles, and bridges chapters, providing semantic refinement and informational expansion.
3. **Narrative Orchestrator** Coordinates the recursive loop, maintains long-term context, evolves keywords, and assembles the final manuscript.

The pipeline operates through a bi-directional feedback loop:

DMLG seed $\rightarrow$ LLM moderation $\rightarrow$ LLM scoring $\rightarrow$ LLM keyword evolution $\rightarrow$ DMLG context update.

This loop repeats for each chapter, enabling multi-generational world evolution.

A structurally critical component of the sequential narrator pipeline is the *chapter transition*. After a chapter is accepted, the evaluator receives both the previous chapter and the newly generated one and must synthesize a coherent narrative bridge. This transition step functions as a dual prompt:

$$(\text{chapter}_{t-1}, \text{chapter}_t) \rightarrow \text{transition}_t.$$

If no coherent transition can be produced—due to drift, motif collision, identity mismatch, or structural inconsistency—the entire chapter is rejected and the pipeline returns to the structural generator for a new candidate. Transition failure is therefore a governance mechanism: it prevents world contamination, enforces canonical continuity, and stabilizes the attractor across generations.

By combining recursive chapter generation with strict transition validation, the pipeline maintains long-horizon coherence and ensures that each accepted chapter fits into the evolving world geometry. This dual-stage recursion—chapter generation followed by transition synthesis—is what enables stable multi-generational narrative evolution within the DMLG-v3 substrate.

7.2 Initialization

The pipeline begins with a narrative configuration specifying chapter length, decoding parameters, grammar constraints, and model size. A structural substrate is loaded with its canonical grammar, page geometry, and curriculum index. The narrative context is initialized as an empty memory that will accumulate actors, motifs, and thematic structure as the book evolves.

Keyword sets are initialized from user-provided parameters and filtered through the curriculum index to ensure canonical grounding.

7.3 Step 1: Canonical Seed Generation

For each chapter, the structural generator produces a canonical seed story based on the current context and keyword set. This involves:

- selection of a sequence embedding guided by keyword relevance,
- grammar and lemma alternation through prefix grammar,
- canonical token generation via lemma-keyed pages,

- decoding through top- k sampling or beam search,
- structural and semantic validation.

The result is a canonical structural draft representing the chapter’s raw form.

7.4 Step 2: Moderation

The evaluator rewrites the canonical seed into natural language, removing artifacts, smoothing semantics, enforcing domain constraints, and ensuring stylistic consistency. This moderated version becomes the chapter’s narrative form.

7.5 Step 3: Title Generation

The evaluator proposes a chapter title. If no meaningful title can be produced, the chapter is rejected.

7.6 Step 4: Scoring and Acceptance

The moderated chapter is scored for coherence, readability, originality, progression, and suitability. Only chapters above a threshold are accepted, ensuring recursive quality control.

7.7 Step 5: Chapter Transition

If a previous chapter exists, the evaluator synthesizes a transition that:

- connects the two chapters,
- maintains thematic consistency,
- avoids introducing new entities,
- preserves narrative flow.

Transitions are validated for coherence and minimal length.

7.8 Step 6: Context Update

Accepted chapters update the long-term narrative context. Canonical tokens are reconstructed, sentences are encoded, and narrative memory integrates new characters, actions, and atmospheric elements. This allows the structural generator to absorb the evolving world.

7.9 Step 7: Keyword Evolution

The evaluator proposes new keywords based on the chapter’s content. These are filtered through the curriculum index and integrated into the next chapter’s generation process, enabling adaptive world evolution.

7.10 Step 8: Ending Validation

Once the minimum chapter count is reached, the evaluator determines whether the current chapter constitutes a valid ending. If so, the book terminates; otherwise generation continues.

7.11 Step 9: Book Assembly

The orchestrator assembles the final manuscript: titles, moderated chapters, transitions, and optional refined variants.

7.12 Structural Interpretation

The sequential narrator pipeline provides:

- **G**: canonical structure, modular pages, transitions,
- **R**: recursive moderation, scoring, context accumulation,
- **I**: keyword evolution, semantic expansion,
- **E**: energetic amplification via the evaluator.

Together with DMLG-v3, the pipeline forms an IEGR-complete hybrid system capable of multi-generational world evolution and emergent narrative structure.

8 Energetic Scaling Through Multi-Variant Generation

While the sequential narrator pipeline already produces coherent long-form narratives through recursive hybrid intelligence, its quality can be further increased by injecting additional energetic capacity (E) into the generative process. Because DMLG-v3 provides a stable geometric-recursive substrate (G, R), and the evaluator layer supplies informational refinement (I), the remaining scaling axis is energetic amplification: allocating more generative effort per chapter and selecting the best candidate.

8.1 Multiple Variants per Chapter

One approach is to generate several candidate chapters for each position in the book. Each candidate undergoes moderation, scoring, and transition validation. The narrator then selects the highest-scoring chapter, optionally using an *early-accept* rule when a candidate exceeds a high threshold. This multi-variant strategy increases narrative quality by exploring a broader semantic and structural space while preserving canonical geometry.

Because transitions act as structural constraints, only candidates that support a coherent bridge to the previous chapter are eligible. This ensures that multi-variant generation improves quality without compromising world fidelity.

8.2 Beam Search Over Chapters

A more powerful extension is *beam search over chapters*. Instead of producing a single linear chain of chapters, the narrator maintains several parallel generative beams. Each beam represents a possible narrative trajectory, with its own sequence of accepted chapters and transitions. After a fixed number of generations, the narrator selects the beam with the highest cumulative score or strongest structural coherence.

Beam search over chapters provides a form of long-horizon energetic scaling: multiple world trajectories are explored in parallel, and only the most coherent trajectory is retained. This method is computationally more expensive but yields significantly higher narrative stability and thematic depth.

8.3 Structural Interpretation

Both multi-variant generation and chapter-level beam search increase the energetic role (E) of the pipeline without altering its geometric or recursive structure. DMLG-v3 continues to supply canonical geometry (G) and recursive depth (R), while the evaluator layer provides informational refinement (I). The pipeline becomes an IEGR-complete system with amplified energetic capacity, allowing it to explore richer narrative possibilities and converge on higher quality attractors.

These strategies illustrate how energetic scaling—not architectural change—is the primary driver for further improvements in narrative quality. They also suggest a path toward more advanced hybrid systems in which multiple beams, variants, or recursive trajectories interact to produce increasingly coherent and expressive long-form narratives.

9 Hybrid IEGR Architecture: DMLG-v3 as a Plastic GR Substrate

The IEGR taxonomy characterizes intelligent systems through four structural roles: energetic efficiency (E), informational richness (I), geometric structure (G), and recursive depth (R). Modern transformer-based LLMs scale E and I aggressively but lack explicit geometric modularity and recursive structure, placing them in the statistical parrot regime. Conversely, DMLG-v3 instantiates G and R strongly but operates with modest energetic and informational capacity, placing it in the canonical parrot regime. When these systems are coupled through a bi-directional narrator pipeline, the resulting hybrid becomes IEGR-complete: all four structural roles are present and functionally integrated.

IEGR-completeness does not imply broad cognitive capability; it simply denotes that the structural prerequisites for coherent long-horizon behavior are present. The magnitude of each role—its scale—determines the system’s practical performance within a given domain. Energetic capacity (E) is particularly important: it governs how deeply and broadly the system can explore its generative space. In this sense, IEGR-complete systems form a structural foundation that can be scaled or layered depending on the demands of the application.

This section describes a minimal hybrid IEGR architecture composed of three components:

1. DMLG-v3: a plastic, trainable geometric-recursive substrate,
2. a transformer-based LLM: an energetic-informational substrate,
3. a bi-directional narrator pipeline: an integrative IEGR layer.

Together, these components form a structurally complete hybrid system capable of long-horizon narrative generation within a bounded domain.

9.1 DMLG-v3 as a Plastic GR Substrate

DMLG-v3 provides the geometric and recursive backbone of the hybrid system. Its structural features—prefix grammar, lemma alternation, deterministic token embeddings, codebook-based output, fixed-size neural architecture, lemma-keyed pages, stable attractors, and functional narrative memory—enable the model to absorb entire fictional universes through curriculum learning.

DMLG-v3 supplies:

- **G (Geometric structure)**: canonical grammar, modular pages, stable attractors, and compositional canonical forms,
- **R (Recursive depth)**: context windows, narrative memory, recursive page updates, and long-horizon coherence.

Because DMLG-v3 is trainable and plastic, it can absorb arbitrary domains through curriculum learning. In the narrative pipeline, it learns the structural rules of a world (e.g., slapstick physics or Victorian gothic atmosphere) and maintains long-term memory across chapters.

9.2 LLM as an IE Substrate

The transformer-based LLM provides the energetic and informational dimensions:

- **E (Energetic efficiency)**: high-throughput inference and large-scale computational capacity,
- **I (Informational richness)**: broad semantic coverage, stylistic flexibility, and world knowledge.

The LLM moderates DMLG-v3 outputs, fixes grammar and semantics, generates chapter titles, scores narrative quality, produces transitions between chapters, and evolves keyword sets. It acts as an informational amplifier for the canonical seeds produced by DMLG-v3.

9.3 The Narrator Pipeline as a Bi-Directional IEGR Layer

The narrator pipeline integrates DMLG-v3 and the LLM through a recursive bi-directional loop:

DMLG generation $\rightarrow$ LLM moderation $\rightarrow$ LLM keywords $\rightarrow$ DMLG context update.

This loop provides:

- additional **G**: structural filtering, canonical consistency, and chapter-level organization,
- additional **R**: recursive moderation, scoring, bridging, and long-horizon narrative planning,
- additional **I**: keyword evolution and semantic expansion.

The pipeline therefore acts as an IEGR integrator: it couples the GR substrate (DMLG-v3) with the IE substrate (LLM) and supplies the missing structural dimensions required for IEGR completeness.

9.4 IEGR Completeness Through Hybrid Composition

The resulting hybrid system can be expressed structurally as:

$$\text{GR}_{\text{DMLG}} + \text{IE}_{\text{LLM}} + \text{iegr}_{\text{pipeline}} = \text{IEGR-complete system}.$$

This architecture is structurally equivalent to a sophisticated parrot: a system in which all four IEGR roles are present, but whose practical capability depends on the scale of each role.

9.5 Energetic Scaling and Performance

IEGR completeness provides the structural foundation for coherent long-horizon behavior. Performance improves as energetic capacity (E) increases:

- larger LLMs increase informational richness (I) and energetic throughput (E),
- larger DMLG models increase geometric structure (G) and recursive depth (R),
- deeper pipelines increase long-horizon planning and world coherence.

Energetic scaling is therefore a practical mechanism for improving narrative quality and structural stability. Multi-variant generation, chapter-level beam search, and recursive refinement are examples of energetic amplification that enhance performance without altering the underlying architecture.

9.6 Conclusion: DMLG-v3 as a Generic GR Substrate

DMLG-v3 provides the geometric and recursive substrate required for IEGR completeness. When coupled with an LLM and a bi-directional narrator pipeline, the resulting hybrid system becomes a structurally coherent cognitive architecture for long-form narrative generation. Energetic scaling enhances its performance, enabling richer world absorption, deeper recursion, and more stable long-horizon behavior. The narrative pipeline thus serves as a concrete, empirically validated example of how IEGR-complete systems can be constructed from complementary parrot regimes.

10 Empirical Results: Autonomous World Absorption and Narrative Evolution

To evaluate the structural claims of the IEGR model and the architectural improvements introduced in DMLG-v3, we conducted a series of empirical experiments using the sequential narrator pipeline. These experiments demonstrate that the hybrid system (DMLG-v3 + LLM + pipeline) is not merely structurally IEGR-complete but also operationally capable of autonomous world absorption, long-horizon narrative coherence, and multi-generational story evolution.

We present results from two independent narrative universes:

1. **Tales of the Dark Forest:** a slapstick ecosystem with four canonical animal attractors,
2. **Tales of the Shrouded City:** a strict Victorian gothic world based on *Dr. Jekyll and Mr. Hyde*.

Both systems were generated fully autonomously, end-to-end, without human intervention in content.

10.1 Tales of the Dark Forest: Emergent Slapstick Ecosystem

Tales of the Dark Forest is the first complete demonstration of the Triadic Cosmos narrator pipeline. Starting from a distilled slapstick curriculum, DMLG-v3 absorbed the structural rules of a chaotic forest environment and developed four stable behavioral attractors:

- **fox:** cunning but overconfident,
- **rabbit:** anxious and reactive,
- **raccoon:** wise but sarcastic,
- **monkey:** chaotic wildcard.

These attractors emerged without explicit agent models. They arose from the interaction between:

- DMLG-v3’s prefix grammar and lemma-keyed pages (G),
- narrative memory and recursive context accumulation (R),
- LLM moderation and keyword evolution (I),
- energetic amplification through the LLM (E).

The resulting ecosystem exhibited:

- **slapstick physics:** unstable ground, falling branches, bubbling lava pits,

- **multi-agent comedic tension:** emergent coordination and conflict,
- **canonical consistency:** stable roles across chapters,
- **long-horizon coherence:** narrative arcs spanning ten chapters.

The book included AI-generated illustrations derived from director’s-cut prompts, demonstrating multimodal consistency across text and imagery. This experiment validated the pipeline’s ability to stabilize micro-model output into coherent long-form narrative.

10.2 Tales of the Shrouded City: Multi-Generational Gothic Evolution

Tales of the Shrouded City is a more demanding test: a strict Victorian gothic universe based on *Dr. Jekyll and Mr. Hyde*. The system was required to maintain:

- canonical characters (Jekyll, Hyde, Utterson, Poole, Bradshaw),
- strict Victorian London atmosphere,
- consistent scientific logic around Jekyll’s experiment,
- thematic focus on internal struggle,
- prohibition of motifs from other gothic works.

The sequential narrator pipeline produced:

- **multiple complete editions,**
- each with **14 combined chapters,**
- each chapter containing a **director’s-cut variant,**
- totaling **42 pipeline chapters per edition,**
- fully autonomous world evolution across generations.

Each edition produced a distinct narrative arc, demonstrating that the hybrid system can:

- absorb a world through curriculum learning (DMLG-v3),
- maintain long-term narrative memory across dozens of chapters,
- evolve keywords and themes recursively,
- generate coherent transitions between chapters,
- detect valid endings through LLM-based structural validation.

The system maintained strict canonical constraints throughout all generations, showing that DMLG-v3’s geometric and recursive substrate can preserve world rules even as the LLM introduces stylistic variation.

10.3 Structural Interpretation of Empirical Results

The empirical results demonstrate that the hybrid system is not merely IEGR-complete in theory but IEGR-functional in practice. The following structural properties were observed:

- **Geometric stability (G)**: prefix grammar, canonical pages, and narrative memory produced stable attractors and consistent world rules.
- **Recursive depth (R)**: context accumulation and multi-step moderation enabled long-horizon coherence across dozens of chapters.
- **Informational expansion (I)**: keyword evolution and LLM moderation enriched the semantic space without breaking canonical structure.
- **Energetic amplification (E)**: transformer inference provided the computational throughput required for stylistic refinement and high-quality narrative generation.

These results confirm the IEGR model’s prediction: hybrid systems composed of complementary parrot regimes can achieve structural completeness and exhibit emergent general-purpose behavior when coupled through a recursive integrator.

10.4 Conclusion

The sequential narrator pipeline demonstrates that DMLG-v3, when combined with a transformer-based LLM and a bi-directional coupling layer, forms a structurally complete IEGR system capable of autonomous world absorption, multi-generational narrative evolution, and long-horizon coherence. Scaling energetic capacity (E) could transform this architecture into human-like AGI.

11 Testable Predictions

The hybrid architecture presented in this paper yields a set of concrete, falsifiable predictions about (i) the behavior of DMLG-v3 as a geometric-recursive substrate, (ii) the dynamics of the sequential narrator pipeline, and (iii) the conditions under which stable long-horizon narrative generation emerges. These predictions can be evaluated empirically using existing systems.

11.1 Predictions About DMLG-v3 as a GR Substrate

Prediction 1: Stable Attractor Formation DMLG-v3 will consistently form stable attractors in any domain with sufficient canonical structure. These attractors will persist even when paired with weaker LLM moderation, indicating that attractor formation is intrinsic to the GR substrate.

Prediction 2: Canonical Constraint Preservation DMLG-v3 will preserve canonical constraints across long narrative horizons. Structural violations will originate primarily from the evaluator layer rather than from DMLG-generated seeds. Replacing the evaluator with a noisy model should not disrupt DMLG’s canonical consistency.

Prediction 3: Curriculum Absorption Given a structured curriculum, DMLG-v3 will absorb domain rules—physics, character roles, thematic constraints—without explicit symbolic encoding. Failure to reproduce these rules under controlled curricula would falsify this prediction.

11.2 Predictions About the Sequential Narrator Pipeline

Prediction 4: Multi-Generational Coherence The pipeline will maintain coherent narrative evolution across many chapters. Using smaller evaluators will degrade quality gradually rather than collapse structure, demonstrating that long-horizon coherence arises from the GR substrate and the recursive pipeline.

Prediction 5: Reliable Ending Detection Ending validation will generalize across domains. If ending detection fails systematically in new narrative styles, the prediction is falsified.

Prediction 6: Keyword-Driven Thematic Progression Keyword evolution will reliably steer the narrative in coherent thematic directions. Removing keyword feedback will produce stagnation or drift, demonstrating its role as an informational driver.

Prediction 7: Transition-Based Structural Governance Chapter transitions will act as an effective structural filter: coherent transitions will correlate strongly with chapter acceptance, and transition failure will reliably prevent world contamination. If transitions fail to predict chapter quality, the prediction is falsified.

11.3 Predictions About Energetic Scaling and Quality

Prediction 8: Smooth Quality Improvement Under Energetic Scaling Increasing energetic capacity—larger evaluators, multi-variant generation, chapter-level beam search—will produce smooth, predictable improvements in coherence, world absorption, and thematic stability. Abrupt discontinuities would falsify this prediction.

Prediction 9: Multi-Variant Selection Improves Stability Generating multiple candidate chapters and selecting the highest-scoring one will consistently outperform single-pass generation. If multi-variant selection fails to improve quality, the prediction is falsified.

Prediction 10: Beam Search Over Chapters Enhances Long-Horizon Structure Beam search over chapters will produce more stable long-range arcs than single-trajectory generation. If chapter-level beams do not outperform single-path pipelines, the prediction is falsified.

11.4 Summary

These predictions focus on structural stability, curriculum absorption, long-horizon coherence, and energetic scaling within the narrative domain. They provide a practical roadmap for evaluating DMLG-v3 and the sequential narrator pipeline as hybrid IEGR systems capable of stable multi-generational narrative generation.

12 Open Questions and Limitations

Although the hybrid architecture presented in this paper demonstrates stable IEGR-complete behavior within narrative domains, several structural limitations and open research questions remain. These limitations do not undermine the empirical results but delineate the boundaries of current systems and identify directions for future work.

12.1 Limitations of the GR Substrate (DMLG-v3)

Informational Scale (I). DMLG-v3 provides strong geometric and recursive structure but remains informationally shallow compared to transformer-based models. Its world absorption capacity depends heavily on curriculum quality and domain structure. Scaling informational richness within DMLG-v3 without compromising determinism or modularity remains an open challenge.

Energetic Throughput (E). DMLG-v3 is intentionally lightweight. This enables interpretability and structural stability but limits raw energetic throughput. Whether future GR substrates can increase energetic capacity while preserving canonical geometry is an open research question.

Domain Dependence. DMLG-v3 performs best in domains with strong canonical structure (e.g., slapstick ecosystems, Victorian gothic fiction). Its behavior in weakly structured or high-entropy domains remains underexplored.

12.2 Limitations of the Sequential Narrator Pipeline

Dependence on the Evaluator Layer. The pipeline relies on the evaluator for moderation, scoring, keyword evolution, and semantic expansion. While DMLG-v3 provides structural stability, the final semantic quality depends on the evaluator’s informational richness. Whether future GR substrates can internalize more semantic refinement is an open question.

Scaling Behavior. The pipeline exhibits smooth capability growth under energetic scaling (larger evaluators, deeper recursion, multi-pass refinement). However, the exact scaling laws for hybrid IEGR systems—linear, sublinear, or superlinear—remain unknown.

Generalization Beyond Narrative Domains. The architecture demonstrates stable behavior in narrative universes. Whether the same IEGR composition generalizes to non-narrative domains such as planning, scientific reasoning, or embodied tasks remains to be tested.

12.3 Open Questions About Energetic Scaling

Energetic Capacity as a Variable Property. A key insight of this work is that energetic capacity is not a fixed threshold but a variable property that can be scaled depending on task demands. This raises several questions:

- What is the minimal energetic capacity required for high-quality performance in specific domains?
- How does energetic efficiency interact with geometric and recursive structure?
- Can hybrid systems dynamically allocate energetic resources across submodules?

IEGR-Scale vs. IEGR-Completeness. IEGR-completeness ensures that all structural roles are present, but practical performance depends on the magnitude of each role. Determining quantitative thresholds for IEGR-scale remains an open theoretical and empirical challenge.

Compositional Hybrid Architectures. The IEGR model suggests that hybrid systems may be compositional: multiple lower-capacity modules could be combined into higher-capacity systems. Whether such compositional architectures outperform monolithic ones remains an open question.

12.4 Open Questions About Biological Parrot Equivalence

Cognitive Profile Matching. Hybrid IEGR-complete systems exhibit structural profiles similar to biological parrots: stable attractors, domain-bound reasoning, recursive planning. Whether these systems can match the full behavioral repertoire of biological parrots (e.g., tool use, social reasoning, meta-learning) remains to be tested.

Scaling Toward Human-Like Performance. It remains unknown whether energetic scaling in IEGR-complete systems produces smooth performance curves, phase transitions, or requires additional structural roles beyond IEGR. Understanding these scaling dynamics is essential for characterizing the upper limits of hybrid architectures.

12.5 Summary

The limitations and open questions identified here outline the frontier of IEGR-based hybrid intelligence. They highlight the need for further research on energetic scaling, domain generalization, compositional architectures, and the quantitative structure of IEGR-scale. The empirical results of the narrator pipeline provide a foundation for addressing these questions, but substantial theoretical and engineering work remains.

13 Conclusion

This paper presented a hybrid generative architecture grounded in the IEGR framework and demonstrated its effectiveness through a fully autonomous narrative system. The IEGR model identifies four structural roles—energetic efficiency (E), informational richness (I), geometric structure (G), and recursive depth (R)—as the foundation for coherent long-horizon behavior. Modern transformer-based models instantiate E and I strongly but lack explicit geometric modularity and recursive structure, while DMLG-v3 provides a robust geometric-recursive backbone with modest energetic and informational scale. Neither architecture alone is sufficient for stable long-form generation, but their complementary integration yields an IEGR-complete system.

DMLG-v3 was introduced as a deterministic GR substrate offering canonical grammar, modular page structure, stable attractors, and long-horizon narrative memory. Coupled with a transformer-based evaluator and integrated through a bi-directional narrator pipeline, the resulting hybrid system exhibits autonomous world absorption, multi-generational narrative evolution, strict canonical constraint enforcement, and stable thematic progression. The pipeline demonstrates that structured recursion, geometric modularity, and energetic amplification can be composed into a coherent long-form generative system that substantially exceeds the capabilities of prompt-driven LLMs.

The empirical results—two fully autonomous books, *Tales of the Dark Forest* and *Tales of the Shrouded City*—provide concrete evidence that hybrid IEGR-complete systems can maintain stable attractors, enforce domain physics, preserve canonical constraints, and generate coherent narrative arcs across dozens of chapters. These results validate the IEGR hypothesis that complementary parrot regimes can be composed into structurally complete generative architectures.

Insight: Structural Scalability of DMLG-v3

A notable implication of this work is that the deterministic structure of DMLG-v3 may enable substantially larger-scale training than earlier GR substrates. Because the output space, page architecture, activation stack, and token geometry are fixed at compile time, the model remains stable even when trained on much larger curricula, broader domains, or deeper page hierarchies.

This opens the possibility of scaling DMLG-v3 to significantly larger parameter spaces and to corpora far beyond single-world narrative datasets.

Moreover, the modular page structure suggests that future versions of DMLG could support thousands of pages, richer lemma families, and more expressive codebooks. If paired with more advanced context frontends—such as hierarchical memory systems, multi-resolution context windows, or structured semantic indexing—the GR substrate could internalize a larger portion of the IEGR loop and approach full IEGR completeness without relying as heavily on external evaluators.

These possibilities remain theoretical, but the empirical stability of DMLG-v3 makes them realistic avenues for future research. The architecture’s deterministic geometry, fixed output space, and stable attractor dynamics provide a foundation for exploring large-scale GR models capable of absorbing complex worlds while maintaining canonical structure and long-horizon coherence.

In summary, this work demonstrates that hybrid IEGR-complete systems can achieve autonomous long-horizon narrative generation, that DMLG-v3 provides a viable geometric-recursive substrate for structured cognition, and that energetic scaling and improved context frontends offer promising directions for advancing the capabilities of modular generative architectures.

A Narrator Pipeline Diagram

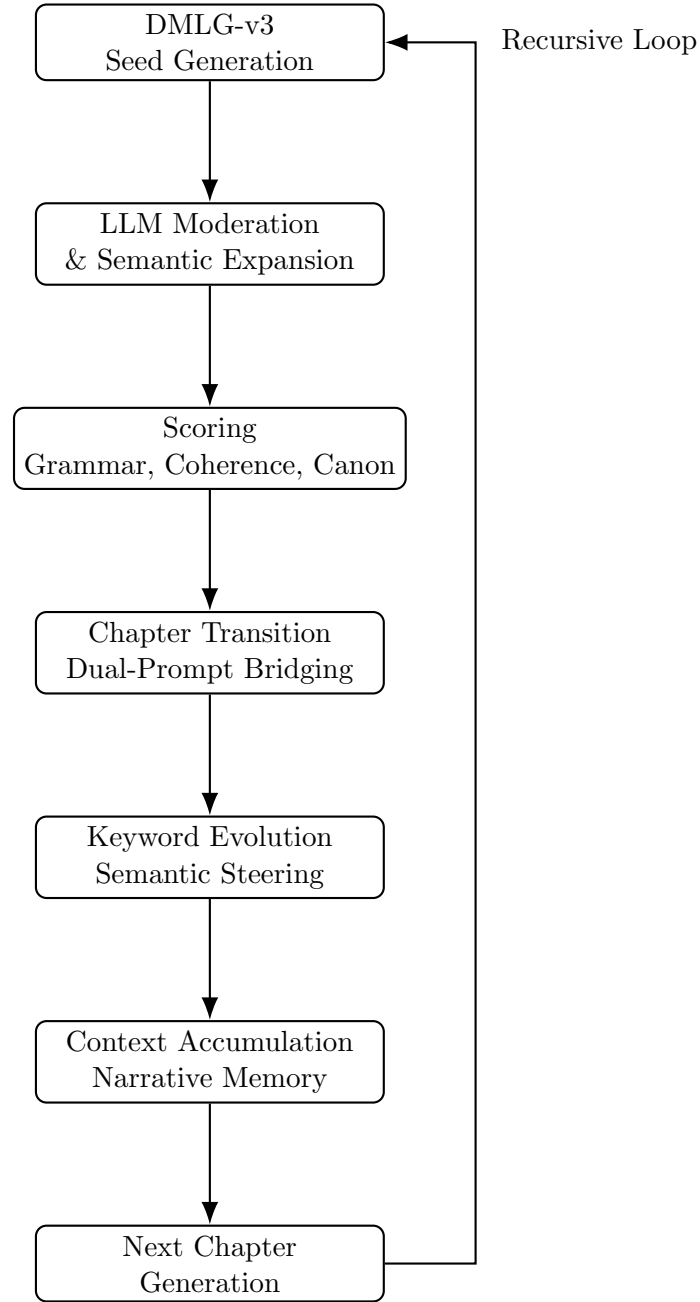


Figure 1: The narrator pipeline: a sequential IEGR loop integrating DMLG-v3 (G–R) with a transformer-based LLM (E–I), including chapter-level transitions.

B IEGR Structural Roles

Role	Definition	Manifestation in the Hybrid System
E (Energetic Efficiency)	Computational throughput, generative bandwidth, and refinement capacity.	Transformer-based evaluator provides high-throughput moderation, scoring, transition synthesis, and keyword evolution. Multi-variant generation and chapter-level beam search amplify energetic capacity.
I (Informational Richness)	Semantic coverage, stylistic flexibility, and world knowledge.	Evaluator supplies semantic expansion, natural-language rewriting, thematic steering, and domain knowledge. Keyword evolution provides controlled informational feedback into the GR substrate.
G (Geometric Structure)	Canonical forms, modularity, attractors, and deterministic update rules.	DMLG-v3 provides prefix grammar, lemma-keyed pages, fixed-size neural architecture, stable attractors, and deterministic token geometry. The narrator pipeline enforces chapter-level structure and canonical constraints.
R (Recursive Depth)	Long-horizon coherence, memory accumulation, and multi-step reasoning.	DMLG-v3 supplies recursive page updates and narrative memory. The sequential narrator pipeline adds recursive moderation, scoring, transition synthesis, and context accumulation across chapters.

Table 1: IEGR roles and their instantiation across DMLG-v3, the evaluator layer, and the narrator pipeline.

C Triadic Cosmos Ecosystem Context

The present paper is part of the broader *Triadic Cosmos* ecosystem: a unified research program that develops a compositional ontology of intelligence together with modular, inspectable computational architectures. The ecosystem spans conceptual theory, executable reference implementations, and fully documented empirical demonstrations, forming a coherent framework for studying hybrid intelligent systems.

All materials, extended notes, and evolving documentation are publicly available in the open GitHub project:

<https://github.com/triadic-cosmos>

Within the `triadic-system` repository, a complete reference implementation of the DMLG-v3 engine, the sequential narrator pipeline, and all supporting modules is provided under the AGPL-3.0 license. The repository includes:

- the full DMLG-v3 engine,
- the narrative memory subsystem,
- the sequential narrator pipeline,
- the trainer and curriculum tools,
- complete runnable examples and reproducible experiments.

Beyond the codebase, the ecosystem also contains fully autonomous narrative demonstrations generated by the hybrid system described in this paper. These demonstrations serve as empirical validation of the IEGR framework and the structural role of DMLG-v3 as a geometric-recursive substrate.

Two example books are currently part of the ecosystem:

1. **Tales of the Dark Forest** A slapstick ecosystem featuring four canonical animal attractors (fox, rabbit, raccoon, monkey), emergent slapstick physics, and multi-agent comedic tension. The book demonstrates how micro-model output can be stabilized into coherent long-form narrative through recursive moderation, keyword evolution, and director’s-cut refinement.
2. **Tales of the Shrouded City** A strict Victorian gothic universe based on *Dr. Jekyll and Mr. Hyde*, generated across multiple editions (42 chapters per edition). The book demonstrates world absorption, long-horizon narrative memory, strict canonical constraint enforcement, and multi-generational story evolution.

The Triadic Cosmos ecosystem therefore provides both the theoretical foundation and the empirical evidence for the claims made in this paper: that hybrid intelligent systems composed of complementary parrot regimes can achieve structural completeness, and that DMLG-v3 serves as a generic GR substrate for IEGR-complete architectures.

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